

Hanna Margrét Pétursdóttir (s253092)

Contents

1	Introduction	4
2	Risk Management Process (NIST SP 800-30)	5
2.1	Prepare for Assessment	5
2.2	Conduct Assessment	5
2.3	Communicate Results	5
2.4	Maintain Assessment	5
3	Ship Description and Asset Identification	6
3.1	Navigation Systems	6
3.1.1	Electronic Chart Display and Information System (ECDIS)	6
3.1.2	Automatic Identification System (AIS)	7
3.1.3	Radar System	7
3.1.4	Global Navigation Satellite System (GNSS/GPS)	7
3.1.5	Integrated Bridge System (IBS)	7
3.1.6	Voyage Data Recorder (VDR)	8
3.2	Operational / Control Systems	8
3.2.1	Engine Control System	8
3.2.2	Ballast System	8
3.2.3	Ballast Water Management System	9
3.2.4	Cargo Management System	9
3.3	Communication and IT Systems	9
3.3.1	Satellite Communication System (SATCOM)	9
3.3.2	GMDSS (Global Maritime Distress and Safety System)	10
3.3.3	Shipboard Network / LAN Infrastructure	10
4	Attacks and Threat Analysis	11
4.1	Common Categories of Cyber Attacks	11
5	Component Risk Analysis	12
5.1	Electronic Chart Display and Information System (ECDIS)	12
5.1.1	Purpose of the Component	12
5.1.2	Threat Sources and Vulnerabilities	12
5.1.3	Impact Analysis	12
5.1.4	Risk Estimation	12
5.1.5	Risk Evaluation	13
5.2	Automatic Identification System (AIS)	13
5.2.1	Purpose of the Component	13
5.2.2	Threat Sources and Vulnerabilities	13
5.2.3	Impact Analysis	13
5.2.4	Risk Estimation	13
5.2.5	Risk Evaluation	13
5.3	Radar System	14
5.3.1	Purpose of the Component	14
5.3.2	Threat Sources and Vulnerabilities	14
5.3.3	Impact Analysis	14
5.3.4	Risk Estimation	14
5.3.5	Risk Evaluation	15
5.4	Global navigation satellite system (GNSS/GPS)	15
5.4.1	Purpose of the Component	15
5.4.2	Threat Sources and Vulnerabilities	15
5.4.3	Impact Analysis	15
5.4.4	Risk Estimation	15
5.4.5	Risk Evaluation	16
5.5	Integrated Bridge System (IBS)	16
5.5.1	Purpose of the Component	16
5.5.2	Threat Sources and Vulnerabilities	16

5.5.3	Impact Analysis	16
5.5.4	Risk Estimation	16
5.5.5	Risk Evaluation	17
5.6	Voyage Data Recorder (VDR)	17
5.6.1	Purpose of the Component	17
5.6.2	Threat Sources and Vulnerabilities	17
5.6.3	Impact Analysis	17
5.6.4	Risk Estimation	17
5.6.5	Risk Evaluation	17
5.7	Engine Control System	18
5.7.1	Purpose of the Component	18
5.7.2	Threat Sources and Vulnerabilities	18
5.7.3	Impact Analysis	18
5.7.4	Risk Estimation	18
5.7.5	Risk Evaluation	18
5.8	Ballast System	18
5.8.1	Purpose of the Component	18
5.8.2	Threat Sources and Vulnerabilities	18
5.8.3	Impact Analysis	18
5.8.4	Risk Estimation	19
5.8.5	Risk Evaluation	19
5.9	Ballast Water Management System	19
5.9.1	Purpose of the Component	19
5.9.2	Threat Sources and Vulnerabilities	19
5.9.3	Impact Analysis	19
5.9.4	Risk Estimation	19
5.9.5	Risk Evaluation	19
5.10	Cargo Management System	20
5.10.1	Purpose of the Component	20
5.10.2	Threat Sources and Vulnerabilities	20
5.10.3	Impact Analysis	20
5.10.4	Risk Estimation	20
5.10.5	Risk Evaluation	20
5.11	Satellite Communication System (SATCOM)	20
5.11.1	Purpose of the Component	20
5.11.2	Threat Sources and Vulnerabilities	20
5.11.3	Impact Analysis	21
5.11.4	Risk Estimation	21
5.11.5	Risk Evaluation	21
5.12	GMDSS (Global Maritime Distress and Safety System)	21
5.12.1	Purpose of the Component	21
5.12.2	Threat Sources and Vulnerabilities	21
5.12.3	Impact Analysis	21
5.12.4	Risk Estimation	22
5.12.5	Risk Evaluation	22
5.13	Shipboard Network / LAN Infrastructure	22
5.13.1	Purpose of the Component	22
5.13.2	Threat Sources and Vulnerabilities	22
5.13.3	Impact Analysis	22
5.13.4	Risk Estimation	23
5.13.5	Risk Evaluation	23
6	Risk Treatment and Communication	24
6.1	Risk Communication Process	24
6.2	Risk Treatment Strategy	24
6.3	Proposed Security Controls (Risk Mitigation)	24
6.4	6.4 Summary of Residual Risk	25

7 Discussion: Maintaining the Assessment	26
8 Conclusion	27
9 Author Contributions	28

1 Introduction

This report looks at cyber risks for the “CyberShip” using the NIST Special Publication 800-30 framework. The NIST framework is used to identify and evaluate risks in a structured way. It makes it easier to find possible threats and weaknesses in the system, as well as the impact these risks could have on the ship. The shipping industry is very important because about 90% of world trade is transported by sea. Today, ships use many digital systems, which makes operations more efficient but also creates new cybersecurity risks.

Modern ships depend on connected systems for navigation, communication, and controlling the ship. Because these systems are connected, they can be targeted by cyber attacks. For example, attackers can send false signals, interfere with GPS, or use malware to gain access to systems. These attacks can cause serious problems, like wrong navigation decisions, accidents, or disruption of ship operations.

In this report, important systems and assets on the CyberShip are identified, and possible threats and weaknesses are analyzed. The risks are then evaluated based on how likely they are to happen and how serious their impact would be. The goal of this report is to understand the main cyber risks and help improve the security of the ship.

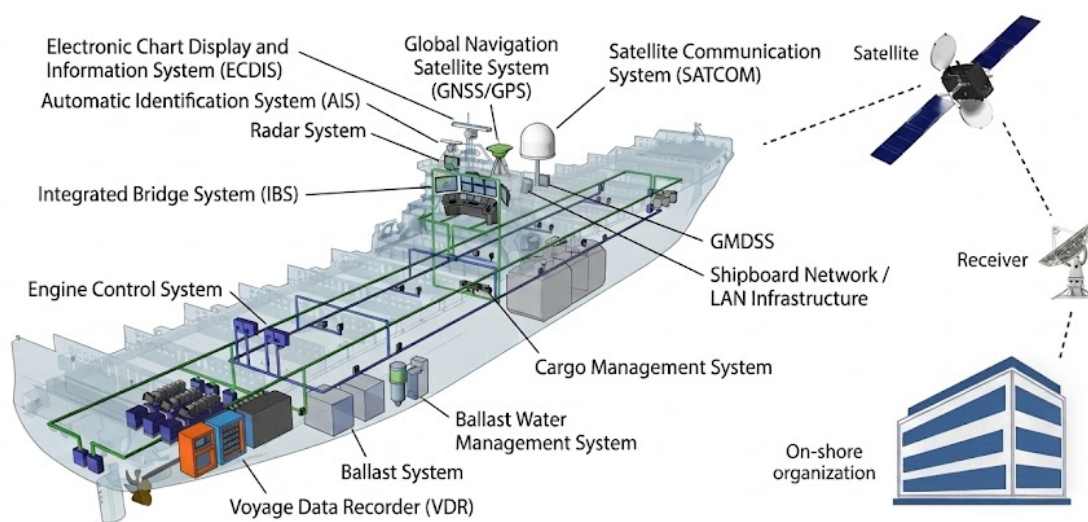


Figure 1: Labelled Components of Cybership

2 Risk Management Process (NIST SP 800-30)

The NIST Special Publication 800-30 framework is a widely used guideline for identifying and managing cybersecurity risks. It provides a structured approach to understanding how threats, vulnerabilities, and system weaknesses can affect an organization or system. It is used to help organizations make better decisions about how to reduce risks and improve security.

In this report, the NIST SP 800-30 framework is used to assess cyber risks on the CyberShip. The framework helps to break down the risk assessment process into clear steps, making it easier to identify important assets, analyze possible threats, and evaluate their impact.

According to NIST SP 800-30 Revision 1, the risk assessment process consists of four primary steps which are followed in this report: [1]

1. **Prepare for Assessment:** Defining the scope and identifying assumptions.
2. **Conduct Assessment:** Identifying threat sources, vulnerabilities, and determining likelihood and impact.
3. **Communicate Results:** Sharing findings with stakeholders to support decision-making.
4. **Maintain Assessment:** Monitoring for changes and evolving threats over time.

2.1 Prepare for Assessment

In this first phase, the scope of the assessment is defined. Critical CyberShip systems such as navigation, communication, and control systems are defined as part of the scope. Assumptions are made regarding the operational environment. Key assets, such as the Integrated Bridge System, Access Control System and Engine System are identified. The goal of this step is to clearly define and understand that is included in the assessment and which assets are important

2.2 Conduct Assessment

In this step we identify possible threat sources, vulnerabilities in the system. Threat sources can include external and inside attackers and vulnerabilities may include outdated software, weak security controls or system interconnectivity making attacks easier to spread. For each identified asset, possible treats are analyzed and the impact is evaluated in terms of safety, financial loss, enviromental damage and system dailures. The combination of likelihood and impact is then used to determine the overall risk level

2.3 Communicate Results

The results of the risk assessment are documented and communicated to relevant stakeholders, such as ship operators and management. Good and effective communication ensures that decision makers understand the severity of risks and can prioritize what security measures should be taken.

2.4 Maintain Assessment

Risk assessment is an ongoing process that must adapt to changing threats and system changes. This includes monitoring the system, updating threat intelligence, and reevaluating risks when new vulnerabilities or technologies are introduced. Regular reviews and updates ensure that the risk management process remains effective and secure.

3 Ship Description and Asset Identification

The “CyberShip” relies on several interconnected cyber-physical systems that support navigation, propulsion, cargo handling, and security operations. These systems are highly integrated, increasing both operational efficiency and cyber risk exposure.

The primary systems are categorized as follows:

- **Integrated Bridge System (IBS):** Includes key navigation components such as Electronic Chart Display and Information System (ECDIS), Automatic Identification System (AIS), Radar, and GNSS/GPS.
- **Propulsion and Machinery Management:** Includes engine control systems and remote monitoring of propulsion and power systems.
- **Cargo Management Systems:** Includes digital cargo manifests, loading/unloading software, and stability management systems.
- **Access Control Systems:** Includes CCTV and Ship Security Alert System (SSAS) used for physical and operational security.

Based on these systems, the following key assets are identified for the risk assessment:

- Electronic Chart Display and Information System (ECDIS)
- Automatic Identification System (AIS)
- Radar System
- Global Navigation Satellite System (GNSS/GPS)
- Integrated Bridge System (IBS)
- Voyage Data Recorder (VDR)
- Engine Control System
- Ballast Water Management System
- Cargo Management System
- Satellite Communication System
- Global Maritime Distress and Safety System (GMDSS)
- Shipboard Network (LAN Infrastructure)

These assets are considered critical as their compromise may lead to navigation errors, loss of propulsion control, cargo damage, or safety incidents affecting crew and environment.

3.1 Navigation Systems

The ships navigation system includes many connected components which work together to support safe route planning and real time navigation on the sea. These systems are in the ships bridge and share data to provide accurate information. Components of this system are ECDIS, AIS, radar and GNSS/GPS. Since these systems are connected and are updated through the internet or with removable media makes these systems vulnerable to cyber attacks. Compromise in one component may affect other components which impacts the safety of the ship due to this each component is analysed individually as part of the navigation system [2][3].

3.1.1 Electronic Chart Display and Information System (ECDIS)

The ECDIS is part of the navigation system its purpose is to display charts and relevant information such as marine charts, navigation data, along with travel information. The ECDIS system uses data from various sensors such as GNSS/GPS gyrocompass and speed sensors. This system is used to plan the ship trip and see it in real time. Some of these systems expand their capabilities by connecting to the radar system, weather system and AIS, along with other systems. Today, ships are required to have this system onboard by international regulations if the ship meets certain requirements [4].

3.1.2 Automatic Identification System (AIS)

The Automatic Identification System provides data to from the ship to shore stations and other ships in real time. The data it transmit includes ships identity, type, position, course, speed, navigational status. AIS relies on GNSS/GPS data and VHF radio communication to broadcast the information automatically, the AIS integrated with other navigation systems onboard like the ECDIS to provide information about of other ships around. If a ship has AIS equipment it needs to be turned on all the time expect where other agreements or rules apply, and the information from the AIS is publicly available which can result in safety concern since everyone can access information sent from the AIS. The system is very capable and can handle over 4.500 reports per minute and can update every two second [5][6].

3.1.3 Radar System

The radar system is essential part of the navigation system to prevent collisions especially in bad weather when visibility is low. Radar system uses radio waves to see other ships and landmasses. Modern ships are often equipped with radar that supports two bands, first is X-band which is a high frequency band and then S-band which is a lower frequency band, the S-band offers better performance in rain and harsher weather condition. These two bands provide the navigation system good understanding of what is surrounding the ship, for an example the navigation system can use information from the radar to show distance from the ground or if there is any floating object around the ship for an example islands, icebergs or rocks. The radar system can identify object all around the ship because it rotates and that way it catches everything around the ship [2][7].

3.1.4 Global Navigation Satellite System (GNSS/GPS)

The Global navigation satellite system is one of the most important part of the navigation system when it comes to precise position for ships. The GNSS/GPS works by using satellites signal that are in orbit around the world to calculate the longitude and latitude of the ship with great accuracy. With this system it is possible to calculate the speed, the course of the ship, with the speed and the course it is possible to calculate the time it takes the ship cover the distance between two points. This system is integrated with the ECDIS, AIS, and radar. Since it relies on signal from satellites this system is vulnerable to interference and manipulation and and since it integrates with other systems on the ship it can impact other system on the ship that depend on accurate position information [2][7][8].

3.1.5 Integrated Bridge System (IBS)

The Integrated Bridge System (IBS) is a central system that connects and manages different navigation devices on the ship, such as ECDIS, AIS, radar, and GNSS/GPS [2, 4]. This allows the crew to see all important navigation information in one place instead of using separate devices. But because all these systems are connected, a problem in one part of the system can affect the others. This also means that if an attacker gains access to the IBS, they may be able to influence multiple systems at the same time [3, 9].

Asset ID	Asset Name	Functional Category	C	I	A
IBS-HW-01	Radar System	Physical / Navigation	Low	High	High
IBS-HW-02	GNSS/GPS Receiver	Physical / Navigation	Low	High	High
IBS-SW-03	ECDIS Software	Logical / Navigation	Medium	High	High
IBS-SW-04	AIS System	Logical / Communication	Low	Medium	High
IBS-IF-05	IBS Interface Console	Human-Machine Interface	Medium	High	High

Table 1: IBS Asset Identification and Sensitivity

3.1.6 Voyage Data Recorder (VDR)

The Voyage Data Recorder (VDR) is used to record important information about the ship during its operation. It collects data such as navigation details, system data, and communication between crew members [10, 11]. The main purpose of the VDR is to understand what happened in case of an accident or incident. It allows investigators to look back at past events and identify the cause of a problem [10]. Even though it is not used for daily decision-making like the IBS, it is still very important because it provides reliable evidence and supports safety improvements.

Asset ID	Asset Name	Functional Category	C	I	A
VDR-HW-01	VDR Recording Unit	Physical / Storage	Low	High	High
VDR-DATA-02	Navigation Data Logs	Information / Data	Medium	High	High
VDR-DATA-03	Audio Communication Records	Information / Data	High	High	Medium
VDR-SW-04	VDR Control Software	Logical / System	Medium	High	High
VDR-IF-05	Data Retrieval Interface	Human-Machine Interface	Medium	Medium	Medium

Table 2: VDR Asset Identification and Sensitivity

3.2 Operational / Control Systems

3.2.1 Engine Control System

The Engine Control System is responsible for managing the ship's propulsion and power generation. It controls key functions such as engine speed, fuel consumption, and overall performance of the propulsion plant [12].

The system supports both local and remote monitoring and control of engine operations, enabling crew and, in some cases, shore-based operators to supervise performance and detect faults [13].

Due to its critical role in vessel propulsion, any malfunction or compromise of this system can directly affect the ship's operability and safety [14].

Asset ID	Asset Name	Functional Category	C	I	A
ENG-HW-01	Engine Control Unit	Physical / Control	Low	High	High
ENG-SW-02	Engine Control Software	Logical / Control	Medium	High	High
ENG-IF-03	Control Interface (HMI)	Human-Machine Interface	Medium	High	Medium

Table 3: Engine Control System Asset Identification and Sensitivity

3.2.2 Ballast System

The Ballast System controls the intake, transfer, and discharge of ballast water to maintain the ship's stability, trim, and structural integrity [13].

By adjusting ballast levels, the system ensures safe operating conditions under varying loading and environmental conditions. It typically relies on pumps, valves, sensors, and automated control mechanisms to regulate water flow [15].

Incorrect operation of this system can lead to instability, excessive structural stress, reduced maneuverability, or safety risks during navigation [14].

Asset ID	Asset Name	Functional Category	C	I	A
BAL-HW-01	Ballast Pumps	Physical / Control	Low	High	High
BAL-HW-02	Valves and Sensors	Physical / Control	Low	High	High
BAL-SW-03	Ballast Control System	Logical / Control	Low	High	High

Table 4: Ballast System Asset Identification and Sensitivity

3.2.3 Ballast Water Management System

The Ballast Water Management System (BWMS) is designed to treat ballast water before discharge in order to prevent the transfer of harmful aquatic organisms and pathogens between different marine environments.

The system typically combines mechanical filtration with disinfection methods such as ultraviolet (UV) radiation or chemical treatment, and operates in conjunction with the ballast system during ballast water intake and discharge processes [13].

Compliance with international regulations, particularly those established by the International Maritime Organization (IMO) under the Ballast Water Management Convention, is a key function of the BWMS. Failure or improper operation of this system may result in environmental damage, legal penalties, and non-compliance with international maritime standards [16].

Asset ID	Asset Name	Functional Category	C	I	A
BWMS-HW-01	Filtration Unit	Physical / Treatment	Low	High	Medium
BWMS-HW-02	UV/Chemical Treatment Unit	Physical / Treatment	Low	High	Medium
BWMS-SW-03	Control System	Logical / Control	Low	High	Medium

Table 5: Ballast Water Management System Asset Identification and Sensitivity

3.2.4 Cargo Management System

The Cargo Management System is used to monitor and manage cargo operations, including loading, unloading, and storage [13].

It maintains digital records such as cargo manifests and may track the condition, quantity, and location of cargo throughout the voyage, often interfacing with other onboard and external systems [15].

The system supports efficient logistics and ensures compliance with safety, regulatory, and handling requirements. Disruptions or errors in this system can lead to cargo damage, operational delays, regulatory violations, or financial losses [14].

Asset ID	Asset Name	Functional Category	C	I	A
CARGO-SW-01	Cargo Management Software	Logical / Business	High	High	Medium
CARGO-DATA-02	Cargo Manifest Data	Information / Data	High	High	Medium
CARGO-IF-03	User Interface System	Human-Machine Interface	Medium	High	Medium

Table 6: Cargo Management System Asset Identification and Sensitivity

3.3 Communication and IT Systems

3.3.1 Satellite Communication System (SATCOM)

The SATCOM system serves as the "Digital Umbilical Cord" for the Cybership. It provides the primary data link for the Vessel Information System (VIS), allowing for real-time synchronization of navigational charts (ENCs), weather overlays, and remote diagnostics. Unlike GMDSS, which is for safety, SATCOM is used for operational efficiency, crew welfare (internet access), and Ship-to-Shore business applications. It typically utilizes Ku-band or Ka-band VSAT for high bandwidth and L-band (Inmarsat/Iridium) as a narrowband backup.

Asset ID	Asset Name	Functional Category	C	I	A
SC-HW-01	VSAT/Inmarsat Antenna	Physical / Comm.	Low	Mod	High
SC-SW-02	SATCOM Modem/Router	Logical / Network	High	High	High
SC-DATA-03	Config & Telemetry	Information / Data	Mod	Mod	Mod

Table 7: SATCOM Asset Identification and Sensitivity

3.3.2 GMDSS (Global Maritime Distress and Safety System)

GMDSS is a globally mandated suite of safety procedures and equipment under the IMO SOLAS Convention. Its primary purpose is to automate distress alerting and ensure that no vessel in distress shall go unheard. It operates independently of the ship's commercial SATCOM. It includes Digital Selective Calling (DSC) for targeted alerts, EPIRBs for location during sinking, and NAVTEX for receiving urgent maritime safety information (MSI).

Asset ID	Asset Name	Functional Category	C	I	A
GM-HW-01	VHF/MF/HF Radio Units	Physical / Life-Safety	Low	High	Vital
GM-HW-02	AIS Transceiver	Physical / Navigation	Low	High	High
GM-DATA-03	MMSI & MSI Data	Information / Safety	Low	Vital	High

Table 8: GMDSS Asset Identification and Sensitivity

3.3.3 Shipboard Network / LAN Infrastructure

The LAN infrastructure is the central nervous system that facilitates the IT/OT Convergence. It utilizes Managed Switches to implement VLAN Segmentation, logically separating the Administrative IT Network (emails, cargo manifests) from the Control OT Network (Engine Monitoring, Bridge Systems). This segmentation is the primary defence against lateral movement of malware.

Asset ID	Asset Name	Functional Category	C	I	A
NW-HW-01	Core/Edge Switches	Physical / Infra.	Low	High	High
NW-SW-02	Firewall / Gateway	Logical / Security	High	High	High
NW-CONF-03	VLAN & ACL Config	Logical / Config.	High	High	High

Table 9: LAN Infrastructure Asset Identification and Sensitivity

4 Attacks and Threat Analysis

Modern ships have many digital components which are highly interconnected, making them exposed to a wide range of cyber threats. These threats can originate from multiple sources such as cybercriminals, crew members, contractors and others.

Ships rely on continuous data exchange between navigation, control, and communication systems. This connectivity improves the operations on the ship, it also increases the attack surface, more systems for an attacker to exploit vulnerabilities and gain unauthorised access to critical systems. The interconnectivity between components can also lead to that a compromise in one system can lead to a compromise in another component [8].

4.1 Common Categories of Cyber Attacks

Cyber attacks that target ship systems can be grouped into several common categories.

Signal-based Attacks

Signal-based attacks target ship system that rely on external signals, for example, the GNSS/GPS. These attacks can include spoofing, where false signals are introduced, or jamming, where signals are completely disrupted. Attacks like these can greatly affect navigation accuracy and reliability [17][18][19].

Data Manipulation Attacks

Data manipulation attacks occur when the attacker can alter or inject false information to the ship systems. An example for an attack like this is manipulation of the AIS data. Data manipulation can have unpredictable outcomes for systems that rely on the data to be correct, such as when the AIS data is incorrect, it can lead to unsafe navigation [20] [21].

Network-based Attacks

Network-based attacks occur when an attacker gains unauthorised access to systems onboard with methods like malware or phishing. Other network attacks can include SQL injection, brute force attack, and exploit weak internet security [8][22].

These common categories of cyber attacks on ships provide an overview of the threat landscape that affects ships systems. The impact of these threats depends on the specific system or component.

In the following section each component or system is analysed and evaluated individually to its threats, vulnerabilities and potential impact.

5 Component Risk Analysis

5.1 Electronic Chart Display and Information System (ECDIS)

5.1.1 Purpose of the Component

As mentioned in section 3.1.1 the electronic chart display and information system is a important component when it comes to the ships onboard navigation system and is used for route planning and real time position. It gathers data from multiple sensors like GNSS/GPS gyrocompass and speed sensors for safe navigation. Due to its central role in navigation, it is classified as a critical asset.

5.1.2 Threat Sources and Vulnerabilities

ECDIS can be exposed to several cyber threats because it is so connected and relies on external data input. One of the primary threats is malware infection, which can happen with the use of an removable media for an example USB device since USB devices are regularly used for chart updates, a malware from the device can compromise the system functionality or change navigation data.

Another threat is GNSS/GPS spoofing where the GPS position data is manipulated or falsified to mask the correct position of the ship, leads to incorrect ships location being displayed which can result in unsafe navigation.

Additionally, unpatched software or outdated software present a threat, since many ECDIS are still being run on Windows XP operating system or older operating systems, these operating systems can have vulnerabilities and me open to cyber attacks. There are ECDIS that have newer operating systems which can have regular updates/patches which tackles this threat [23][24][25].

5.1.3 Impact Analysis

Since the system primarily processes navigational data which is not very sensitive information but unauthorised access might still pose operational risk the confidentiality is of moderate importance.

The integrity is critical. If the ECDIS or positioning data is manipulated it could lead to incorrect navigation planning which can result in major consequences for an example collision or ground or navigate the ship to a location the attacker wants it to go to.

Availability is also essential, since if the ECDIS functionality is lost or disrupted it would significantly reduce the situational awareness and the ability to safely plan navigation.

5.1.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
IBS-SW-03	Malware infection	High	High	High
IBS-HW-02	GNSS/GPS spoofing	High	High	High
IBS-SW-03	Unpatched software	Medium	High	High

Table 10: Risk Estimation for Electronic Chart Display and Information System

Since the ECDIS is updated with a use of removable media Malware infection is assessed as high likelihood, and a malware can disrupt the system functionality or be used to alter important navigation data the impact is set as high.

The ECDIS relies on external data positioning then manipulating position information may lead to wrong ships location being displayed which can result in unsafe navigation therefore GNSS/GPS spoofing is considered as high likelihood and high impact.

Unpatched or outdated software is assessed as medium likelihood since many but not all ECDIS operate on old operating systems then the likelihood depends on if the operating system is still supported for security updates. The impact is set to high since if the system is exploited it can affect the systems integrity and availability.

The risk estimation was driven by integrity and availability as these were the most important parts of the CIA analysis.

5.1.5 Risk Evaluation

The ECDIS is critical part of the ships navigation system and the results of the risk estimation[[section link](#)] indicate a high risk. Multiple threats were identified like malware infection, GNSS/GPS spoofing which both have high likelihood and high impact leading to increased risk with the system.

The most important concerns is integrity and availability of the system which are essential to navigate safely, as previously mentioned any disruption or manipulation of the system can lead to incorrect route or the ships positioning which increase the risk of collision or grounding. Because of these potential consequences the identified risk are not acceptable and effort should be put into focusing on reducing these threats. ECDIS represents as a high risk component.

5.2 Automatic Identification System (AIS)

5.2.1 Purpose of the Component

As mentioned in section 3.1.2 the purpose of the Automatic identification system is to exchange real time information between ships and station on land. It transmits data like ships identity, position, speed and the course of the ship. AIS relies on GNSS/GPS data for position and uses the VHF radio to communicate, it broadcasts the information automatically. The AIS is a part of other navigation systems including the ECDIS, it makes it capable to monitor surrounding traffic and helps to make good navigation decisions. Due to its role, providing real time ships information, AIS is considered an critical asset within the navigation system

5.2.2 Threat Sources and Vulnerabilities

Since AIS designed as an open and unencrypted system it is vulnerable to several cyber threats. One of the most critical threat is AIS spoofing, it is where false ships information is transmitted to the system, which can result in appearance of an ship that does not exist or incorrect positioning data on the ECDIS.

Another attack is AIS message tampering, which is when an AIS message is intercepted and modified and then send out this could make it appear that the ship was somewhere it was not. These attacks are possible because of the lack of authentication mechanisms which makes AIS vulnerable to unauthorised transmission which allows attacks to send forged messages or alter real messages from other ships [20][21].

5.2.3 Impact Analysis

In the AIS confidentiality is low since the AIS data is open not encrypted and is intentionally broadcasted and publicly accessible for safety. The integrity is critical, since if the data is manipulated or false data is broadcasted it may lead to incorrect ships position, identities or movements. This can affect navigation and increase the risk of collision. Availability is also critical since if there are any disruption or loss of AIS signals it can affect safety especially in low visibility where quick information about ships nearby is essential. Therefore the primary elements at risk in AIS are integrity and availability of data that is transmitted because the impact navigation safety.

5.2.4 Risk Estimation

The risk estimation for the AIS is based on the identified threats AIS spoofing, message tampering, and unauthorised transmission. AIS spoofing is assessed as high likelihood because the system is open and unencrypted which allows any attacker to transmit false vessel information, the impact of this is high as spoofed data may result in unsafe navigation. AIS message tampering is also considered as high likelihood and high impact since the AIS message can be intercepted and altered for an example attacker may alter the identity of the ship, position data. Unauthorised transmission is assessed as high likelihood also because of the lack of encryption which allows attackers to send out forged messages into the system or alter legitimate transmission, but the impact is assessed as medium since the radar can help detect some inconsistencies. The risk estimation was driven by integrity and availability since these were the most important parts of the CIA analysis.

5.2.5 Risk Evaluation

The automatic identification system is a high risk component due to its design as an open and unencrypted system, the identified threats like AIS spoofing and message tampering affect the safety and therefore

Asset ID	Threat Event	Likelihood	Impact	Risk Score
IBS-SW-04	AIS spoofing	High	High	High
IBS-SW-04	Message tampering	High	High	High
IBS-SW-04	Unauthorised transmission	High	Medium	High

Table 11: Risk Estimation for Automatic Identification System

increase the risk. The most important concern is the data integrity since wrong data can lead to wrong information about ships around and can lead to unsafe navigation or collision. Even though some wrong data can be detected by other navigation components for an example the radar or the ECDIS, the availability of the real time information from the AIS is critical for navigation so the identified risks cannot be considered acceptable and actions to improve the detection of wrong or falsified data should be taken.

5.3 Radar System

5.3.1 Purpose of the Component

As mention in section 3.1.3 is an essential part of the ships navigation system to detect objects, ships, and landmasses especially in low visibility condition and uses radio waves to accomplish that. It integrates with other navigation system like the ECDIS and AIS and can be used to cross referenced the data to support safe navigation. Because of the role of the Radar system is is considered as an critical asset within the ships navigation system.

5.3.2 Threat Sources and Vulnerabilities

The radar system is vulnerable to several cyber threats. One threat is that it is possible to send commands via the radar to the radar system by sending radar signals in a specific pattern it is called the open door, but the system already has to have malware on it for this to work, but if the system has the malware then it is possible to send commands via the radar signal to the system, this works even if the system is air gapped meaning it is not connected to the internet. Another threat is radar spoofing. Radar spoofing is he manipulation of radar signals to essentially create false targets or echos in the radar display. This can lead to appearance of ships or obstacle that are not there. Radar jamming is also a threat, it involves the disruption of radar signals this can affect or completely block the systems ability to detect ships or objects which can affect safety during low visibility conditions [20][26].

5.3.3 Impact Analysis

Like the other navigation components confidentiality is not of great importance, since the radar data is not considered sensitive data. But the integrity is critical because the manipulation of radar signals with methods like radar spoofing or using the open door attack it can lead to false ships or objects being detected which can affect the navigation of the ship. Availability is also very important with attacks like radar jamming it can disrupt the system ability to detect other ships or object which can be dangerous in low visibility environments where radar is one of the primary source of information. Then the primary elements at risk are the integrity and availability of the radar.

5.3.4 Risk Estimation

The risk estimation for the radar system is based on the identified threats, open door attack, radar spoofing, and radar jamming. The open door attack is assessed as low likelihood but high impact, since it needs prior malware infection in the system, but if the system is compromised then attackers can send commands through radar signals even if the system is air gapped which is a major security concern. Radar spoofing is assessed as medium likelihood but high impact, it requires technical knowledge and capabilities to manipulate radar signals if that is done then it can introduce false ships or object which leads to wrong interpretation of surroundings of the ship. Radar jamming is also assessed medium likelihood but high impact, it involves the disruption of radar signals which can disrupt or completely block the system ability to detect ships or object around this can be dangerous during low visibility conditions where radar is essential. This risk estimation was driven by integrity and availability since those were the most important parts of the CIA analysis.

Asset ID	Threat Event	Likelihood	Impact	Risk Score
IBS-HW-01	Open door attack	Low	High	Medium
IBS-HW-01	Radar spoofing	Medium	High	High
IBS-HW-01	Radar jamming	Medium	High	High

Table 12: Risk Estimation for Radar System

5.3.5 Risk Evaluation

The radar system is a high risk component because of its role in safe navigation. Though some threats like the open door attack have lower likelihood since it needs the system to already to be compromised the impact remains significant. The most important concern is the compromise of integrity and the systems availability. Radar spoofing can make it look like there are false ships or objects around the ship which can affect navigation decisions, and radar jamming can disrupt or completely block the systems ability to detect ships or object around it. Both attacks can lead to unsafe conditions especially in low visibility environment or conditions where the radar system servers as an important part of the information. Some capability is to cross check radar data with other systems such as AIS or ECDIS but then the question rises which data to trust because of the conflicting information. Because of the high impact of these threats the identified risks cannot be considered acceptable without mitigation measures, and the radar system is assessed as an high risk component as part of the whole navigation system due to its direct impact on navigation.

5.4 Global navigation satellite system (GNSS/GPS)

5.4.1 Purpose of the Component

As mentioned in section 3.1.4 the purpose of the global navigation satellite system is to provide precise position for ships it operates by using signal from multiple satellites to calculate the longitude and latitude to provide the precise position. This system is an important part of the ship navigation system and it is integrated with the ECDIS, AIS and radar. It is one of the primary source of position information and is then critical for route planning and safe ship navigation. This system is classified as an critical asset.

5.4.2 Threat Sources and Vulnerabilities

The global navigation satellite system is vulnerable to several cyber threats since it depends on signal from satellites. One of the threats is GNSS/GPS spoofing, this attack works by an attacker replicates the satellite signals with an identical signal that is strong enough to force out the original transmission, this works because the GNSS receivers lock on to the strongest available signal, when the attacker has the spoof signal in place the false transmission can confuse the navigation systems about calculations for location, velocity and heading. Another attack is GNSS/GPS jamming it is when the satellites signal is disrupted which results in loss of positioning capability. Meaconing is also a threat to the GNSS/GPS system, which is an attack which involves to rebroadcast legitimate GNSS/GPS signals but with a delay, which can also mislead position calculations [18][17][19].

5.4.3 Impact Analysis

Like with the other components confidentiality is of low importance as the GNSS signals are publicly accessible and not considered sensitive. The integrity is critical as manipulated positioning data like with spoofing are meaconing can result in incorrect calculations of the ship position, velocity, and heading. Availability is also very important as GNSS jamming or signal disruption can get in the way of the system to receive positioning data and not receiving positioning data will affect navigation and affect multiple systems within the navigation system like ECDIS and AIS. The primary elements at risk are the integrity and availability of GNSS.

5.4.4 Risk Estimation

The risk estimation for GNSS/GPS system is based on the identified threats of spoofing, jamming, meaconing. GNSS/GPS spoofing is assessed as high likelihood and high impact, since an attacker can transmit false signals that can override the correct transmissions which can result in incorrect positioning

data which is being used by multiples systems within the navigation system. GNSS/GPS jamming is also assessed as high likelihood and high impact since it does not take much to be able to disrupt satellite signals and prevent the receiver from getting the positioning data, this will affect navigation capabilities and affect other system that depend on this information. Meaconing is assessed as medium likelihood but has high impact, it requires the attacker to capture and retransmit legitimate signals, but it can still mislead positioning calculations which disrupts the navigation system. This risk estimation was driven by integrity and availability since those were the most important parts of the CIA analysis.

Asset ID	Threat Event	Likelihood	Impact	Risk Score
IBS-HW-02	GNSS/GPS spoofing	High	High	High
IBS-HW-02	GNSS/GPS jamming	High	High	High
IBS-HW-02	Meaconing	Medium	High	High

Table 13: Risk Estimation for Global navigation satellite system

5.4.5 Risk Evaluation

The global navigation satellite system (GNSS/GPS) is a high risk since its critical role is the primary source of positioning data for multiple systems within the navigation system. The known attacks that were identified like spoofing, jamming, and meaconing are both high likelihood and have high impact. The most important concern is the integrity and the availability of the system since the attacks can lead to signal disruption can cause complete loss of navigation capability which greatly increase the risk of collision or grounding, also it can be used for piracy of the ship [17].

Also because other systems within the navigation system depend on data from this that an attack on the GNSS/GPS system can affect multiple systems like ECDIS and AIS. Given this the identified risks cannot be considered acceptable without proper mitigation. The GNSS/GPS is classified as a critical asset within the navigation system due to its importance and affect on other systems.

5.5 Integrated Bridge System (IBS)

5.5.1 Purpose of the Component

The purpose of the IBS is to bring together different navigation systems into one interface. This makes it easier for the crew to monitor the ship and make decisions. Instead of checking multiple devices, they can rely on the IBS to provide a complete overview of the ship's situation.

Because it is used all the time during navigation, the IBS is critical for safe and efficient operation of the ship.

5.5.2 Threat Sources and Vulnerabilities

The IBS can be targeted by different types of cyber attacks, including GPS spoofing and jamming, AIS spoofing, malware attacks, and unauthorized access. AIS systems are particularly vulnerable because they lack encryption and authentication, making it possible to send false information [27].

GPS spoofing attacks can mislead navigation systems by providing incorrect position data, which may cause ships to deviate from their intended route [28]. These types of attacks have been observed in real-world maritime environments and are becoming more frequent [29, 30]. These attacks can affect the accuracy and reliability of the information shown to the crew.

5.5.3 Impact Analysis

The IBS has several important elements that can be affected by cyber attacks. These include Navigation data (such as position, speed, and route), sensor data from GPS, radar, and AIS, system availability and communication between connected systems. If any of these elements are compromised, it can directly affect the safety of the ship. [31].

5.5.4 Risk Estimation

Using the NIST qualitative scale:

Asset ID	Threat Event	Likelihood	Impact	Risk Score
IBS-HW-02	GPS Spoofing	High	High	High
IBS-HW-02	GPS Jamming	Medium	High	High
IBS-HW-04	AIS Spoofing	High	Medium	High
IBS-IF-05	Malware via USB	Medium	High	High
IBS-IF-05	Unauthorized Access	Medium	High	High

Table 14: Risk Estimation for IBS using NIST qualitative scale

5.5.5 Risk Evaluation

High Priority: The IBS is a critical system for navigation and is constantly in use. Even small errors in navigation data can lead to serious accidents such as collisions or grounding.

Risks are only acceptable with strong mitigation measures, including secure system configuration, network segmentation, access control, and continuous monitoring.

5.6 Voyage Data Recorder (VDR)

5.6.1 Purpose of the Component

The VDR records important ship data, including navigation information, sensor data, and communications. Its main purpose is to support accident investigations and provide evidence of what happened during a voyage.

5.6.2 Threat Sources and Vulnerabilities

Possible attacks on the VDR include deletion or modification of recorded data, unauthorized access to sensitive information, and disabling or interrupting the recording process. These types of threats are common in maritime and industrial control systems, where attackers may try to manipulate or hide critical data [32, 33].

5.6.3 Impact Analysis

The main elements at risk in the VDR are the integrity of recorded data, the availability of data, and the confidentiality of communication records. If these are compromised, it can affect investigations, legal accountability, and operational decision-making [33]. In some cases, system disruption or data loss may prevent proper analysis of incidents [34].

5.6.4 Risk Estimation

Using the NIST qualitative scale:

Asset ID	Threat Event	Likelihood	Impact	Risk Score
VDR-DATA-02	Data Deletion	Medium	High	High
VDR-DATA-02	Data Modification	Medium	High	High
VDR-DATA-03	Unauthorized Access	Medium	Medium	Medium
VDR-SW-04	System Disruption	Low	High	Medium

Table 15: Risk Estimation for VDR using NIST qualitative scale

5.6.5 Risk Evaluation

High Priority: The VDR is essential for accident investigation and accountability. If data is lost or modified, it becomes very difficult to understand what happened during an incident.

Risks are acceptable only with mitigation measures such as data integrity protection, secure storage, access control, and regular system checks.

5.7 Engine Control System

5.7.1 Purpose of the Component

The Engine Control System manages the ship's propulsion, including engine speed, power output, and overall mechanical performance. It supports both local and remote monitoring of engine operations and is critical for maintaining vessel movement and safe ship operation [12, 13].

5.7.2 Threat Sources and Vulnerabilities

Threat sources include external attackers exploiting remote access systems, insiders with privileged access, and malware introduced through connected networks. Key vulnerabilities include weak authentication, outdated software, and connectivity between IT and operational systems [12, 13, 14].

5.7.3 Impact Analysis

- **Safety impact:** Loss of propulsion may lead to dangerous situations at sea.
- **Operational impact:** Complete disruption of vessel movement.
- **Environmental impact:** Increased risk of collision or drifting incidents.
- **Financial impact:** High repair costs and operational losses.

5.7.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
ENG-SW-02	Unauthorized Remote Access	Medium	High	High
ENG-SW-02	Malware Infection	Medium	High	High
ENG-IF-03	Insider Misuse	Low	High	Medium

Table 16: Risk Estimation for Engine Control System

5.7.5 Risk Evaluation

Critical Priority: The Engine Control System is essential for propulsion and vessel safety. Even threats with moderate likelihood result in high overall risk because of the severe operational and safety consequences. Strong access control, network segmentation, and continuous monitoring are therefore required.

5.8 Ballast System

5.8.1 Purpose of the Component

The Ballast System controls the intake, transfer, and discharge of water to maintain the ship's stability, trim, and balance during operation. It is therefore an important control system for safe vessel operation [13, 15].

5.8.2 Threat Sources and Vulnerabilities

Threat sources include system misuse, external attacks, and sensor manipulation. Key vulnerabilities include reliance on automated control systems, sensor data, and limited cybersecurity protections in industrial control systems [13, 15, 14].

5.8.3 Impact Analysis

- **Safety impact:** Incorrect ballast levels can destabilize the vessel.
- **Operational impact:** Reduced maneuverability and performance.
- **Environmental impact:** Risk of improper discharge affecting ecosystems.
- **Financial impact:** Potential fines and operational losses.

5.8.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
BAL-HW-02	Sensor Manipulation	Medium	High	High
BAL-SW-03	System Misconfiguration	Medium	Medium	Medium
BAL-SW-03	Unauthorized Access	Low	High	Medium

Table 17: Risk Estimation for Ballast System

5.8.5 Risk Evaluation

High Priority: Although the system is generally less exposed than standard IT systems, incorrect ballast control can directly affect vessel stability and safety. The risks are acceptable only with mitigation measures such as sensor validation, monitoring, and protection of the control logic.

5.9 Ballast Water Management System

5.9.1 Purpose of the Component

The Ballast Water Management System (BWMS) is responsible for treating ballast water before discharge in order to prevent the transfer of harmful aquatic organisms between marine environments. It operates alongside the ballast system and supports compliance with international environmental regulations, particularly those defined by the International Maritime Organization (IMO) under the Ballast Water Management Convention [16].

5.9.2 Threat Sources and Vulnerabilities

Threat sources include external attackers, insider misuse, and malware introduced through connected systems.

The BWMS relies on automated control systems and sensors, making it part of the ship's operational technology environment [13]. As a result, it may be vulnerable to threats such as control logic manipulation, data tampering, and unauthorized access.

Additionally, increasing digitalization and system integration introduce further cyber risks to such systems [14].

5.9.3 Impact Analysis

- **Environmental impact:** Improper treatment may lead to ecological damage through the transfer of invasive species.
- **Regulatory impact:** Non-compliance with IMO requirements may result in fines or legal consequences.
- **Operational impact:** Disruption of ballast operations may affect vessel stability procedures.
- **Financial impact:** Costs related to delays, penalties, and system repair.

5.9.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
BWMS-HW-02	Treatment Failure / Manipulation	Low	High	Medium
BWMS-SW-03	Control System Attack	Medium	High	High
BWMS-SW-03	Data Manipulation	Medium	Medium	Medium

Table 18: Risk Estimation for Ballast Water Management System

5.9.5 Risk Evaluation

High Priority: Although the likelihood of some attacks is moderate, the environmental and regulatory consequences can be significant. The risk is acceptable only with mitigation measures, including strong

monitoring, system validation, and protection of the control system.

5.10 Cargo Management System

5.10.1 Purpose of the Component

The Cargo Management System is used to monitor and manage cargo operations, including loading, unloading, storage, and cargo-related data. It supports logistics, regulatory compliance, and the safe handling of cargo throughout the voyage [13, 15].

5.10.2 Threat Sources and Vulnerabilities

Threat sources include cybercriminals targeting business systems, insiders, and malware attacks. Key vulnerabilities include integration with IT networks, remote access, and insufficient access control. Because cargo systems form part of the ship's operational technology and business environment, they are exposed to both operational and information-security risks. [13, 15, 14].

5.10.3 Impact Analysis

- **Operational impact:** Disruption of cargo handling processes.
- **Financial impact:** Loss or damage to goods and contractual penalties.
- **Security impact:** Exposure of sensitive cargo data.
- **Reputational impact:** Loss of customer trust.

5.10.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
CARGO-SW-01	Ransomware Attack	High	High	High
CARGO-DATA-02	Data Breach	Medium	High	High
CARGO-IF-03	Unauthorized Access	Medium	Medium	Medium

Table 19: Risk Estimation for Cargo Management System

5.10.5 Risk Evaluation

High Priority: The Cargo Management System is highly exposed because of its integration with IT networks and business processes. Cyberattacks such as ransomware and data breaches can cause significant operational and financial damage. Strong access control, network security, and regular system updates are therefore required.

5.11 Satellite Communication System (SATCOM)

5.11.1 Purpose of the Component

As mentioned in section 3.4.1, the Satellite Communication System (SATCOM) serves as the "Digital Umbilical Cord" for the ship. Its primary purpose is to provide the high-bandwidth data link necessary for essential Ship-to-Shore communications. This includes facilitating the Vessel Information System (VIS), enabling the real-time synchronization of navigational charts (ENCs), updating weather overlays, and allowing for remote maintenance and diagnostics. Furthermore, it utilizes Ku-band, Ka-band, and L-band frequencies to support business applications and crew welfare through internet access. Because it is vital for both operational efficiency and continuous shore-side connectivity, SATCOM is classified as a critical operational asset. [3]

5.11.2 Threat Sources and Vulnerabilities

Signal Jamming/Interference: Physical layer denial of service.

- **Man-in-the-Middle (MitM):** Intercepting unencrypted L-band or VSAT traffic.

- **Terminal Compromise:** Exploiting unpatched firmware or default credentials on the Below Deck Unit (BDU).

5.11.3 Impact Analysis

- **Operational impact:** Loss or disruption of SATCOM connectivity interrupts the Vessel Information System (VIS), preventing the real-time synchronization of navigational charts (ENCs), weather overlays, and remote diagnostics.
- **Security impact:** Man-in-the-Middle (MitM) attacks or terminal compromise (exploiting the Below Deck Unit) could expose sensitive ship-to-shore business applications and crew data.
- **Financial impact:** Disruption of high-bandwidth operational efficiency tools and delayed business communications can result in logistical delays and associated monetary losses.

5.11.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
SC-DATA-03	Credential Theft	High	Medium	Medium
SC-SW-02	Data Interception	Medium	High	High
SC-HW-01	Signal Jamming	Low	High	Medium

Table 20: Risk Estimation for SATCOM

5.11.5 Risk Evaluation

High Priority: While jamming is localized, the constant internet exposure makes "Terminal Compromise" a persistent threat. Mitigation via multi-factor authentication (MFA) and firmware hardening is mandatory.

5.12 GMDSS (Global Maritime Distress and Safety System)

5.12.1 Purpose of the Component

The Global Maritime Distress and Safety System (GMDSS) is a globally mandated suite of radio and satellite equipment required under the IMO SOLAS Convention. Its primary purpose is to automate distress alerting and ensure reliable coordination for Search and Rescue (SAR) operations, guaranteeing that a vessel in distress will always be heard. It operates entirely independently of the ship's commercial SATCOM and utilizes specialized equipment like Digital Selective Calling (DSC) for targeted alerts, EPIRBs for location tracking during emergencies, and NAVTEX for receiving urgent maritime safety information (MSI). Due to its direct role in preserving the "Safety of Life at Sea," the GMDSS is considered an absolutely vital, life-safety asset.

5.12.2 Threat Sources and Vulnerabilities

- **AIS Spoofing:** Broadcasting false vessel positions or "Ghost Ships" to disrupt navigation.
- **DSC False Alerts:** Triggering automated distress calls to exhaust SAR resources or mask criminal activity.
- **Replay Attacks:** Recording and re-broadcasting safety messages to cause confusion.

5.12.3 Impact Analysis

- **Safety impact:** Because GMDSS is mandated under the SOLAS Convention, any disruption directly threatens the "Safety of Life at Sea" by disabling automated distress alerting and preventing unheard vessels from receiving help.
- **Operational impact:** Triggering DSC false alerts or broadcasting "Ghost Ships" via AIS spoofing can cause immense confusion, disrupt normal navigation, and unnecessarily exhaust Search and Rescue (SAR) resources.

- **Security impact:** Replay attacks that record and re-broadcast safety messages can be used maliciously to mask criminal activity or misdirect authorities during an emergency.

5.12.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
GM-HW-02	AIS Spoofing	Medium	High	High
GM-HW-01	False Distress Alert	Medium	Extreme	High
GM-DATA-03	Radio DoS	Low	Extreme	Medium

Table 21: Risk Estimation for GMDSS

5.12.5 Risk Evaluation

Critical/Safety Priority: Because GMDSS directly impacts “Safety of Life at Sea” (SOLAS), any risk to integrity is evaluated as unacceptable. Although likelihood is lower due to specialized RF requirements, the impact is catastrophic.

5.13 Shipboard Network / LAN Infrastructure

5.13.1 Purpose of the Component

The Shipboard Network or LAN Infrastructure acts as the central nervous system of the ship. Its primary purpose is to serve as the connectivity backbone that enables essential data exchange between various zones, including the Bridge, Engine Room, and Administrative offices. By facilitating IT/OT convergence, it allows modern digital systems to operate efficiently. The infrastructure relies on managed switches to implement strict VLAN segmentation, logically separating the administrative IT network (handling emails and cargo manifests) from the critical operational technology (OT) network (controlling engine monitoring and bridge systems). Because it underpins the communication of nearly all other digital systems onboard, a failure here could lead to a total vessel blackout, making it a highly critical infrastructure asset.

5.13.2 Threat Sources and Vulnerabilities

Ransomware/Lateral Movement: Moving from the Crew/Admin VLAN to the Control VLAN.

ARP Poisoning: Intercepting internal traffic within a subnet.

VLAN Hopping: Bypassing logical segmentation to access restricted OT segments.

5.13.3 Impact Analysis

- **Operational impact:** A compromised network backbone halts the essential data exchange between the Bridge, Engine Room, and Administrative offices, which can rapidly lead to a total vessel blackout.
- **Security impact:** Techniques such as VLAN hopping and ARP poisoning allow attackers to bypass logical segmentation, enabling ransomware or malicious actors to move laterally across the network.
- **Safety impact:** If lateral movement progresses from the Administrative IT Network to the Control OT Network (e.g., Engine Monitoring), attackers gain the ability to manipulate physical systems, directly endangering the crew and the vessel’s structural integrity.

5.13.4 Risk Estimation

Asset ID	Threat Event	Likelihood	Impact	Risk Score
NW-HW-01	Ransomware Movement	High	High	Critical
NW-SW-02	Unauthorized Access	High	Medium	High
NW-CONF-03	VLAN Hopping	Medium	High	High

Table 22: Risk Estimation for Network and LAN Infrastructure

5.13.5 Risk Evaluation

Critical Priority: The LAN is the most vulnerable point due to the variety of end-points (crew devices, vendor laptops). A failure in network segmentation (NW-CONF-03) could lead to a total vessel blackout. Immediate implementation of "Zero Trust" principles at the hardware level is required.

6 Risk Treatment and Communication

6.1 Risk Communication Process

In alignment with Step 3 of the NIST SP 800-30 framework, the findings of this risk assessment must be communicated effectively to all relevant stakeholders to support informed decision-making and resource allocation. The primary audience for this communication includes the Company Security Officer (CSO), the Master (Captain), shore-side IT/OT security teams, and the ship owners. [1]

To ensure the risks are understood and actionable:

- **Integration:** The identified vulnerabilities and required controls must be formally integrated into the vessel's mandatory Ship Security Plan (SSP) to ensure crew awareness and operational compliance.
- **Reporting Mechanisms:** Executive summaries focusing on business and safety impacts (e.g., potential loss of propulsion, safety to life) will be provided to management. Detailed technical specifications and mitigation requirements will be routed directly to the engineering and IT staff.
- **Frequency:** Risk communication is an ongoing process. The risk profile and subsequent communication will be updated during annual security audits, following major shipboard network upgrades, or immediately after a cybersecurity incident

6.2 Risk Treatment Strategy

To address the vulnerabilities identified in the component risk analysis (Section 5), a structured risk treatment strategy is required. The primary approach applied to the critical and high-risk systems on the CyberShip is Risk Mitigation, which involves implementing administrative, technical, and physical controls to reduce the likelihood or impact of a threat event to an acceptable level.

For certain legacy Operational Technology (OT) components where patching is technologically infeasible, a combination of Risk Acceptance (with enhanced continuous monitoring) and Risk Avoidance (disabling unused, vulnerable features) will be employed. [35, 36]

6.3 Proposed Security Controls (Risk Mitigation)

Based on the threat analysis and risk evaluation, the following specific technical and administrative controls are proposed to mitigate the highest priority risks across the vessel's systems: [37, 38]

Table 23: Proposed Risk Treatment Plan for CyberShip Systems

Component	Primary Threat	Proposed Controls	NIST Category
Shipboard LAN	Lateral Movement / Ransomware	Implement strict VLAN segmentation between IT and OT networks. Deploy Zero Trust architecture at the hardware level.	Preventive
ECDIS & IBS	USB Malware Infection	Disable physical USB ports. Enforce strict clean-ship policies using dedicated, scanned USB drives.	Preventive
GNSS / GPS & AIS	Signal Spoofing / Jamming	Deploy anti-spoofing antennas and redundant sensors. Cross-reference GPS with radar and AIS.	Detective
SATCOM	Credential Theft	Enforce Multi-Factor Authentication (MFA) and maintain rigorous firmware patching schedules.	Preventive

Continued on next page

Table 23 – Continued from previous page

Component	Primary Threat	Proposed Controls	NIST Category
Engine Control	Unauthorized Access	Secure remote diagnostics via encrypted VPNs. Implement robust access control and anomaly monitoring.	Preventive
Cargo Mgmt.	Ransomware / Data Breach	Isolate servers from public internet. Implement automated, offline immutable backups.	Preventive
Ballast Systems	Sensor Manipulation	Implement sensor validation algorithms to cross-check levels against physical draft readings.	Detective

6.4 6.4 Summary of Residual Risk

The implementation of the controls outlined above is designed to bring the vessel's overall cybersecurity posture within an acceptable risk tolerance. The table below illustrates the anticipated reduction in risk levels from the initial assessment to the residual risk remaining after controls are applied. [39, 40]

Table 24: Summary of Residual Risk Following Control Implementation

Component	Initial Risk	Primary Mitigation Strategy	Residual Risk
ECDIS / IBS	High	Port disabling and strict USB update protocols.	Medium
GNSS / GPS	High	Hardware redundancy and cross-sensor validation.	Medium
Engine Control	High	Network segmentation and VPN-only remote access.	Low-Medium
Cargo Management	High	Network isolation and immutable offline backups.	Low-Medium
Ballast Systems	High	Sensor validation and logical access controls.	Medium
SATCOM	High	MFA, firmware hardening, and L-band monitoring.	Medium
Shipboard LAN	Critical	Zero Trust ACLs and strict VLAN segmentation.	Medium
VDR	Medium-High	Immutable storage and detailed access logging.	Low
GMDSS	Medium	Procedural training for false-alert identification.	Low

7 Discussion: Maintaining the Assessment

The shipping industry faces a major challenge when it comes to cybersecurity because many systems on ships are used for a very long time. Ship systems can stay in operation for 10-20 years and are not updated frequently like modern IT systems. These systems are often called "legacy systems". Over time these systems can become outdated and may no longer support modern security features. This makes them easier targets for cyber attacks.

Maintaining a NIST risk assesment is something that must be done continuously. New threats are always being discovered and attackers find new ways to exploit systems. This means that the risk assessment must be updated regularly in hand with these new threats.

An important part of maintaining a risk assessment is regular testing of systems. These tests help identify issues and threats before they can be exploited by attackers.

As much as it is important to keep the system safe, it is also important to keep crew members informed and educated on safety protocols. Many cyber incidents happen because of simple mistakes made by people. Crew members should receive ongoing cybersecurity training. They should understand common treats and how to follow safe practices when using systems aboard the ship.

Monitoring is another key part of maintaining the assessment. Systems need to be continuously monitored for unusual activity. This can include checking logs, network traffic and system performance. If something can be detected early it's possible a cyber attack can be stopped before causing damage. Without monitoring attacks can go unnoticed for a long time, porbibly increasing the damage done.

Maintaining a risk assessment requires good planning and organization. There should be clear procedures for updating systems, incident responses and risk reviews. Roles and responsibilities should be clearly defined so that everyone knows what to do in the case of a cyber incident.

8 Conclusion

This report applied the NIST SP 800-30 risk assessment framework to analyze cybersecurity risks in the CyberShip environment. The assessment followed a structured process consisting of preparation, risk analysis, communication of results, and continuous maintenance.

By identifying critical assets across navigation, bridge, operational, and communication systems, and analyzing relevant threats and vulnerabilities, the report established a comprehensive understanding of the CyberShip's risk landscape. The component-level analysis showed that several systems, particularly navigation systems such as ECDIS, AIS, and GNSS/GPS, as well as operational control systems including the Engine Control System and Ballast Systems, exhibit high-risk profiles due to their critical role in safe vessel operation.

The analysis highlighted realistic cyber threats, including spoofing, malware, unauthorized access, and data manipulation. A key finding is the high level of interconnectivity between systems, where shared data and network infrastructure allow attacks on one component to propagate across multiple systems, increasing both likelihood and impact.

Based on the risk evaluation, several risks were classified as high or critical, requiring mitigation measures such as access control, network segmentation, continuous monitoring, regular software updates, and validation of critical sensor data.

Furthermore, the assessment emphasizes that cybersecurity in maritime environments is not a one-time activity but an ongoing process. Continuous monitoring, effective risk communication, and regular reassessment are essential to address evolving threats and system changes.

In conclusion, the CyberShip represents a complex cyber-physical system in which cybersecurity is directly linked to safety, operational reliability, and environmental protection. The application of a structured and continuous risk management approach is therefore essential to ensure resilient and secure maritime operations.

9 Author Contributions

The work in this report was divided among the group members as follows:

- **Birkir Freyr Konráðsson:** Responsible for Attacks and Threat Analysis (Section 4) and Navigation Systems (Section 3.1 and Section 5.1–5.4), including ECDIS, AIS, Radar, and GNSS/GPS.
- **Brynja Eyfjörð Jónsdóttir:** Responsible for Bridge and Monitoring Systems (Section 3.2 and Section 5.5–5.6), including IBS and VDR. Also responsible for the Introduction (Section 1), Risk Management Process (Section 2), and Discussion: Maintaining the Assessment (Section 7).
- **Hanna Margrét Pétursdóttir:** Responsible for Operational and Control Systems (Section 3.3 and Section 5.7–5.10), including Engine Control System, Ballast System, Ballast Water Management System, and Cargo Management System. Also responsible for the Conclusion (Section 8).
- **Pinac Joshi:** Responsible for Communication and IT Systems (Section 3.4.1–3.4.3 and Section 5.11–5.13), including Satellite Communication System (SATCOM), GMDSS, and Shipboard Network / LAN Infrastructure. Also responsible for Risk Treatment and Communication (Section 6). Made CyberShip Labelled Components Diagram Figure 1.
- **All members:** Contributed to discussion, review, and final editing of the report.

Generative AI

Generative AI was used while working on this project to help further our understanding of topics as well as structuring and refining the report. AI was also used to generate image1 in the introduction.

References

- [1] National Institute of Standards and Technology, “Guide for conducting risk assessments (nist sp 800-30 revision 1),” tech. rep., NIST, 2012.
- [2] C. Hays, “15 types of navigation equipment used on modern ships,” 2025. Accessed: 27 March 2026.
- [3] BIMCO, “The guidelines on cyber security onboard ships,” 2016. Accessed: 27 March 2026.
- [4] Raymarine, “What is ecdis?,” 2025. Accessed: 27 March 2026.
- [5] International Maritime Organization, “Ais transponders,” n.d. Accessed: 27 March 2026.
- [6] U. C. Guard, “Automatic identification system (ais) overview,” n.d. Accessed: 27 March 2026.
- [7] R. Alexander, “Types of navigation equipment used on modern ships, an overview!!!,” 2024. Accessed: 27 March 2026.
- [8] European Union Agency for Cybersecurity (ENISA), “Port cybersecurity: Good practices for cybersecurity in the maritime sector,” 2019. Accessed: 27 March 2026.
- [9] European Union Agency for Cybersecurity (ENISA), “Cyber security aspects in the maritime sector,” tech. rep., 2011.
- [10] I. M. Organization, “Voyage data recorder (vdr),” 2020. Accessed: 2026.
- [11] M. Insight, “What is voyage data recorder (vdr)?,” 2024. Accessed: 2026.
- [12] CR Classification Society, “Guidelines for cyber security onboard ships,” 2025. Accessed: 2026.
- [13] SUTD iTrust, “Guidelines for cyber risk management in shipboard operational technology systems,” 2022. Accessed: 2026.
- [14] International Maritime Organization, “Guidelines on maritime cyber risk management,” 2025. Accessed: 2026.
- [15] International Chamber of Shipping, “Guidelines on cyber security onboard ships,” 2024. Accessed: 2026.
- [16] International Maritime Organization, “Ballast water management convention,” n.d. Accessed: 2026.
- [17] C. Lo, “Gps spoofing: What’s the risk for ship navigation?,” 2019. Accessed: 27 March 2026.
- [18] A. Androjna and M. Perkovič, “Impact of spoofing of navigation systems on maritime situational awareness,” *Transactions on Maritime Science*, vol. 10, no. 2, pp. 361–373, 2021.
- [19] A. Androjna and M. Perkovic, “Gnss vulnerabilities vs cyber challenges in maritime navigation,” 2024.
- [20] W. C. Leite Junior, C. C. de Moraes, C. E. P. de Albuquerque, R. C. S. Machado, and A. O. de Sá, “A triggering mechanism for cyber-attacks in naval sensors and systems,” *Sensors*, vol. 21, no. 9, p. 3195, 2021.
- [21] Center for International Maritime Security (CIMSEC), “Security or safety: What is ais really for?,” n.d. Accessed: 27 March 2026.
- [22] BIMCO, “The guidelines on cyber security onboard ships,” 2024. Version 5.
- [23] International Hydrographic Organization (IHO), “The enx and ecdis cyber security instruction,” 2023. Accessed: 27 March 2026.
- [24] NorthStandard, “Ecdis cyber security,” 2018. Accessed: 27 March 2026.
- [25] Y. B. Kurt, E. Uflaz, E. Akyuz, R. E. Kurt, and O. Turan, “Analysis of human reliability in detecting gps spoofing on ecdis in congested waterways under evidential reasoning and heart approach,” *Computers Security*, vol. 151, p. 104316, 2025.
- [26] D. Pradko, “Tech note: Cybersecurity in embedded and sensor systems,” 2022. Accessed: 27 March 2026.
- [27] K. Omholt-Jensen, “Signal wars at sea: Ais and gps jamming and spoofing explained,” 2026.

- [28] AI at Sea, “Gps spoofing detection for maritime,” 2025. Accessed: 2026.
- [29] T. Marine, “Safeguarding against gnss attacks at sea,” 2026. Accessed: 2026.
- [30] W. P. Organization, “Ais spoofing and jamming in the persian gulf,” 2025. Accessed: 2026.
- [31] C. . Security, “Cyber-attacks on ship navigation systems,” 2026.
- [32] European Union Agency for Cybersecurity (ENISA), “Port cybersecurity: Good practices,” tech. rep., 2019.
- [33] K. Tam and K. Jones, “Cyber-risk assessment for maritime systems,” *WMU Journal of Maritime Affairs*, 2019.
- [34] CISA, “Industrial control systems cybersecurity,” 2023. Accessed: 2026.
- [35] SUTD iTrust, “Guidelines for cyber risk management in shipboard operational technology systems,” tech. rep., Singapore University of Technology and Design, 2022.
- [36] BIMCO, “The guidelines on cyber security onboard ships,” tech. rep., BIMCO, 2021.
- [37] International Association of Classification Societies, “Ur e26: Cyber resilience of ships,” tech. rep., IACS, 2022.
- [38] Cybersecurity and Infrastructure Security Agency, “Industrial control systems cybersecurity,” tech. rep., CISA, 2023.
- [39] National Institute of Standards and Technology, “Guide for conducting risk assessments,” Tech. Rep. NIST Special Publication (SP) 800-30 Rev 1, U.S. Department of Commerce, 2012.
- [40] International Organization for Standardization, “Iso/iec 27005:2022 - information security, cybersecurity and privacy protection — guidance on managing information security risks,” 2022.